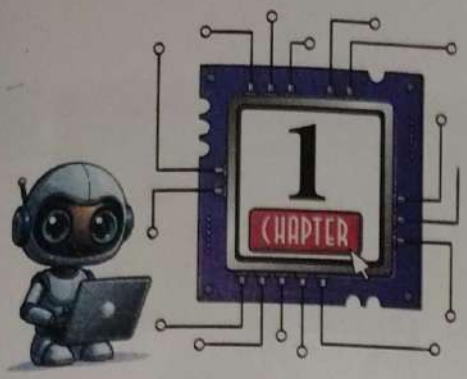


The Super Machine : Computer

Let's Recall

Identify the pictures and tick (✓) the machines.





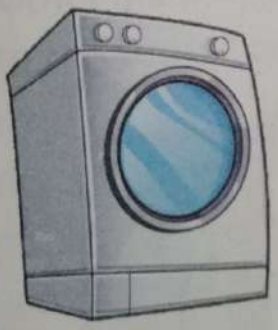

In this Chapter



- What is a Machine?
- Is Computer a Machine?
- Types of Machines
- Types of Computers

MACHINE

Machine is a man-made thing. It makes our life easier and faster. It helps save our time and energy. Machines are built to do specific tasks.



A **Washing machine** is used to wash clothes at home.

An **iron** is a machine which is used to press our clothes.



An **air-conditioner** is a machine used to cool our homes in the summer.





A **refrigerator** keeps the fruits and vegetables fresh.

INFO Byte

Machines can be very small, like a watch, and also very big, like a truck.

Activity Corner

Write 'O' for machines found outside the house and 'I' for machines found inside the house.













TYPES OF MACHINES

Let us learn about some more type of machines :



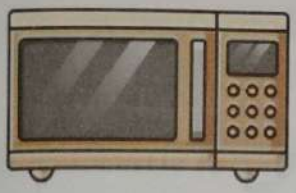
Bicycle

Some machines work with our efforts.



Car

Some machines run on petrol or diesel.



Microwave

Some machines run on electricity.



Wall clock

Some machines run on batteries.

What is this? It looks like a TV, but it has more parts than a TV.



COMPUTER - A SMART MACHINE

This is a machine called computer.

Other machines help us do one or two things only. A computer helps us to do many things. It helps us solve sums, draw pictures, listen to music, play games and more. That is why a computer is called a smart machine.



Let's Get Better

Why is it important to use machines, like computers, carefully and safely?

TYPES OF COMPUTERS

Computers can generally be classified by size :

Desktop Computer



A computer kept on a desk or a table is called a desktop computer. It is also called a personal computer. We cannot carry it from one place to another.

We can play games, draw pictures, watch videos and learn new things on a desktop computer. It is used in homes and offices. We need to plug it into electricity for it to work.



Laptop Computer

A Laptop computer is smaller than a desktop computer. It is about the size of a notebook. We can easily carry it from one place to another. Despite its small size, it can perform all the tasks that a desktop computer can do. We do not need to keep it plugged into electricity, as it works on a battery. It is used in offices, homes, meetings and more.

Other Types of Computers :



Smart Phone



Smart Watch



Tablet



I. Tick (✓) the correct option :

1. Which of the following is a machine?

(a)  ☐ (b)  ☐ (c)  ☐

2. Which of the following machines need electricity?

(a)  ☐ (b)  ☐ (c)  ☐

3. Which of the following machines need our efforts to work ?

(a)  ☐ (b)  ☐ (c)  ☐

4. Which of the following is a Laptop?

(a)  ☐ (b)  ☐ (c)  ☐

II. Match the following :

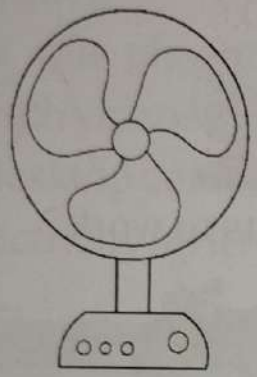
Objects	Description
1. 	(a) Laptop
2. 	(b) Give power to remote and watch
3. 	(c) Desktop Computer
4. 	(d) Work on diesel

III. Fill in the blanks with the help of the box :

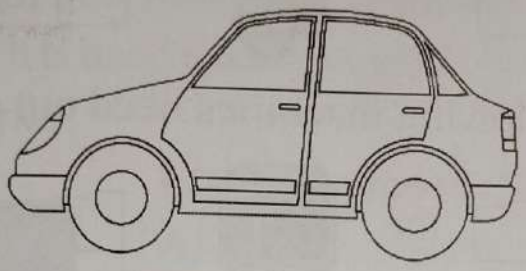
Tablet
Machine
Monitor
Computer

1. A ..... makes our lives easier and faster.
2. A ..... looks like a television screen.
3. A ..... helps us to solve sums, draw pictures and play games.
4. A ..... is a type of computer.

IV. Fill in the missing letters and colour the machines :



F ☐ N



☐ A R



☐ A T ☐ H

Lab Activity

Visit the computer lab and see how computer looks like.

Rules to be followed :

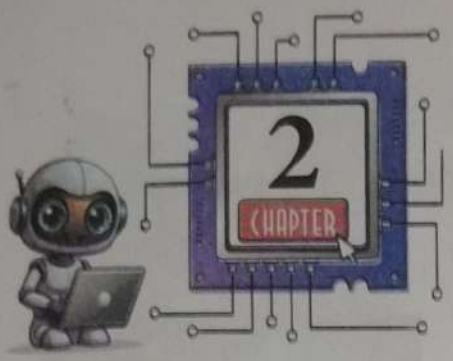
- Listen to teacher and follow the directions.
- Make a queue and move to the computer lab.
- Be gentle with computers and equipment.

Activity : Tick (✓) the things that you see in the lab.

1. Tables ☐ 2. Tablets ☐ 3. Chairs ☐
4. Fan ☐ 5. Laptops ☐ 6. Desktop Computers ☐

FOR THE TEACHER

Take the students to the computer lab and show them what a computer looks like.



Parts of a Computer

Let's Recall

Write the name of the machines you see below :





In this Chapter



- Parts of a Computer
- Mouse
- Monitor
- Printer
- CPU
- Speaker
- Keyboard
- Headphone

We have learnt that computer is a machine that works on electricity. It is very useful for us. Today, it is an important part of our life. In this chapter, we will know about the different parts of the computer.



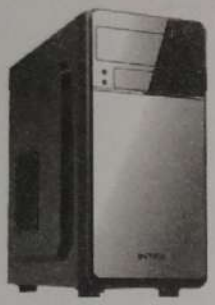
PARTS OF A COMPUTER

Our body is made up of different parts. Just like our body, computer is also made up of different parts, they are :

Monitor



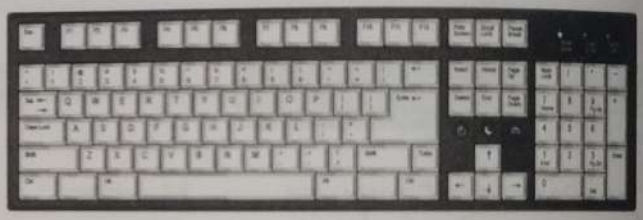
It is the display screen of the computer. It looks like a television. It shows the work we have done on the computer. We can see text, pictures, cartoons, movies and more on the monitor.



CPU

CPU is the brain of the computer. It stands for Central Processing Unit. It controls all the other parts of the computer. All the other parts are connected to CPU through wires.

Keyboard

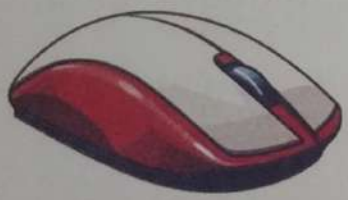


A keyboard has many buttons on it called keys. The keys are used to write on the computer by typing. Numbers, letters and other special symbols are present on the keyboard.

INFO Byte

There are 104 keys on the keyboard.

Mouse



A mouse is a small device connected to the computer through wire. It is used to select things on the computer system. We can draw pictures and point at things on the computer by mouse.

Printer



A printer is a machine. Whatever we see on the monitor, the printer puts it on paper. This is called printing. So, a printer prints the pictures and words on a paper. The printed paper is called printout.



Speaker

A speaker is a device which helps us listen to recorded voices. We listen to music on the computer with the help of speakers. They are connected using wires.

Headphone

Headphones are also used to listen to the music on the computer.



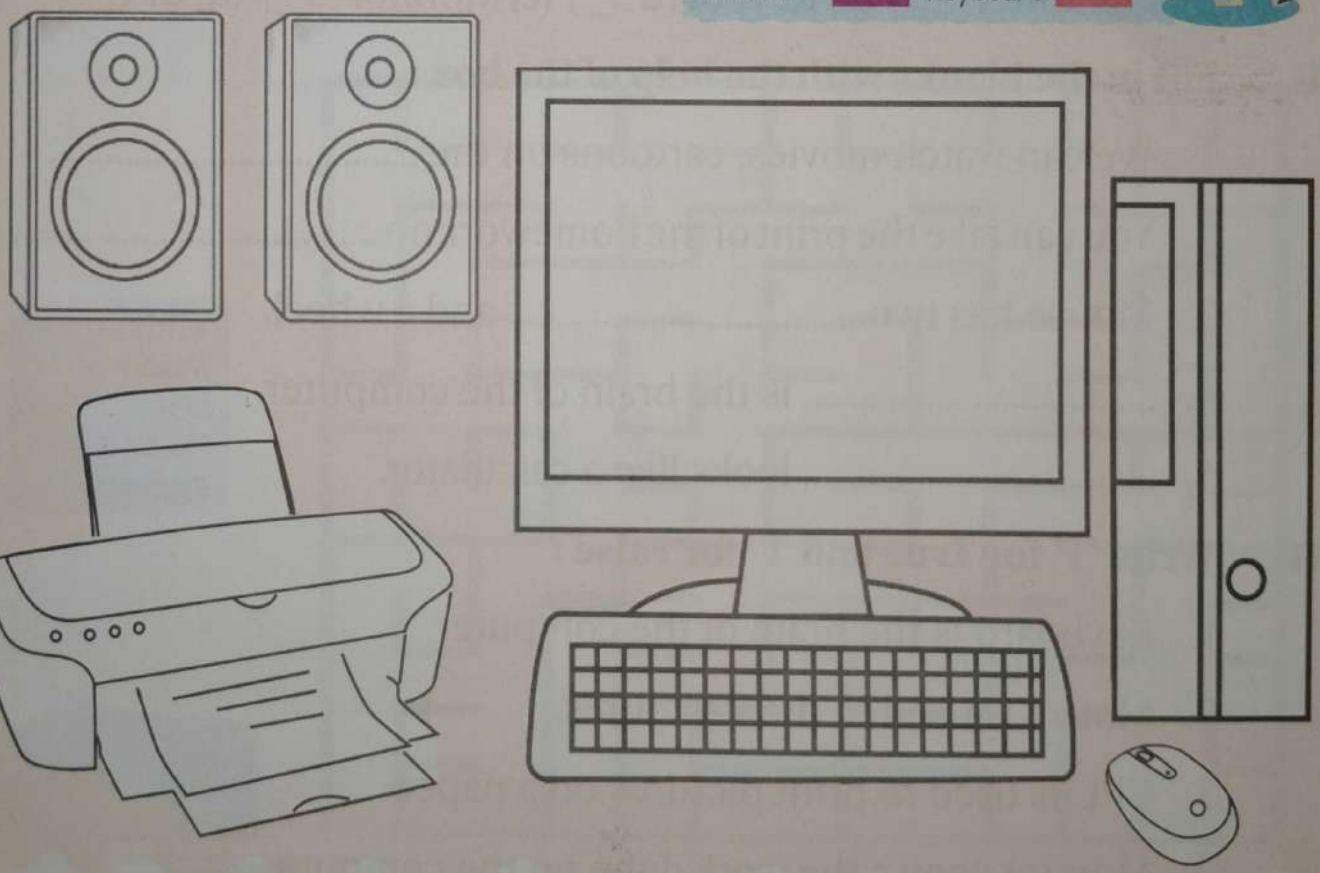
Let's Get Better

Why is it important to take care of all the parts of a computer and use them gently?

Activity Corner

Colour the various parts of a computer.

Speakers		CPU	
Monitor		Mouse	
Printer		Keyboard	





I. Tick (✓) the correct option :

1. I have different keys for typing.
   (a) Printer ☐ (b) Keyboard ☐ (c) Monitor ☐ (d) CPU ☐
2. I look like a television and help people watch movies and cartoons.
   (a) Printer ☐ (b) Keyboard ☐ (c) Monitor ☐ (d) CPU ☐
3. I am the brain of the computer and all parts of the computer are connected to me.
   (a) Printer ☐ (b) Monitor ☐ (c) CPU ☐ (d) Keyboard ☐
4. I help print the pictures for your homework.
   (a) Printer ☐ (b) Keyboard ☐ (c) Monitor ☐ (d) CPU ☐

II. Fill in the blanks with the help of the box :

1. We can watch movies, cartoons on the ..... .
2. You can take the print of the homework on ..... .
3. Mouse has two ..... and a wheel.
4. ..... is the brain of the computer.
5. ..... looks like a calculator.

Buttons
Monitor
CPU
Printer
Keyboard

III. Write 'T' for True and 'F' for False :

1. Keyboard is the brain of the computer. ☐
2. Mouse is used to draw pictures. ☐
3. CPU is used to print pictures on a paper. ☐
4. Monitor shows the work done on the computer. ☐

IV. Complete the names of the following parts of a computer :

1. M___SE
2. K__YB___RD
3. M__N__T__R
4. PR__NT__R

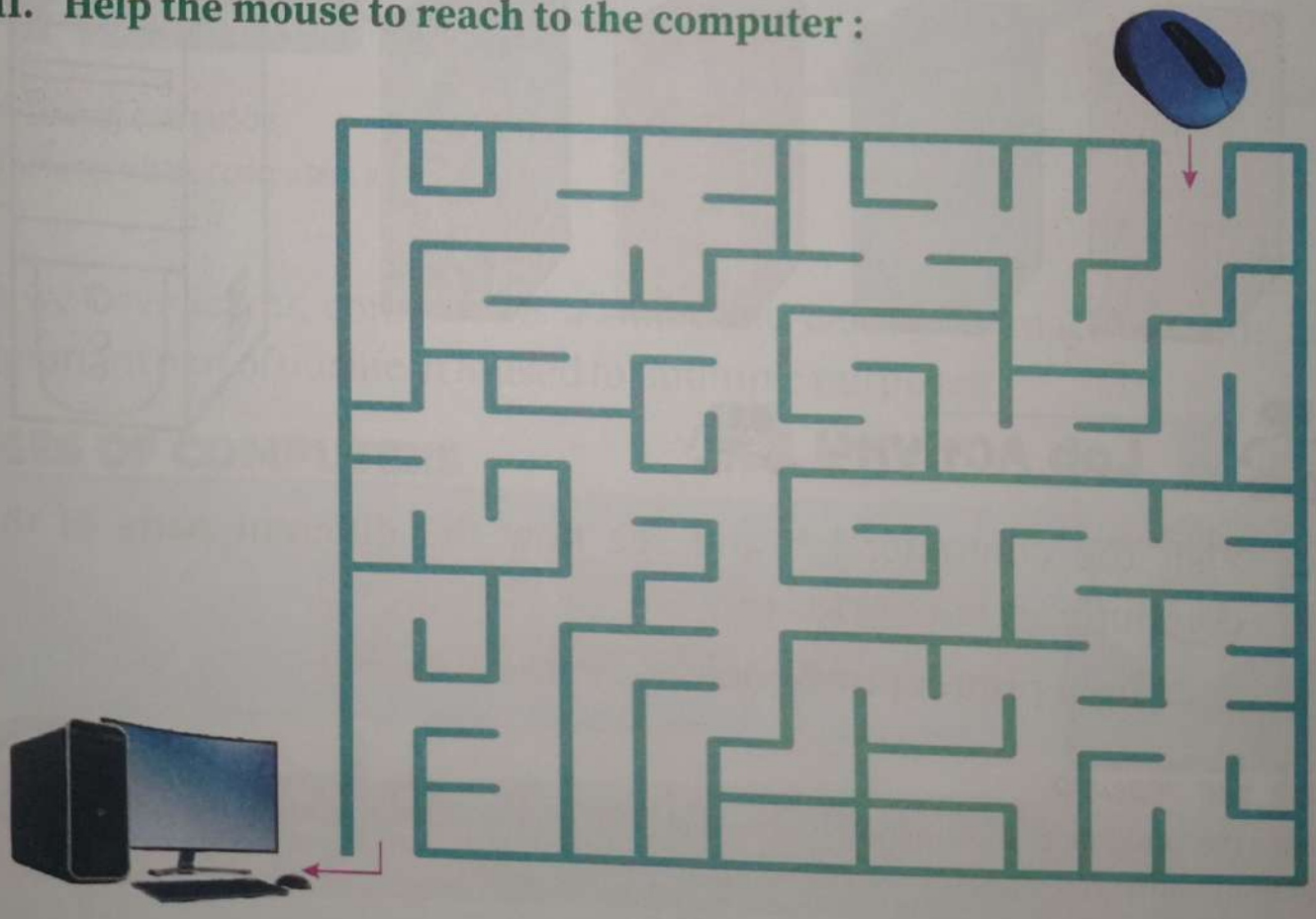
V. Identify the parts of the computer and circle them :

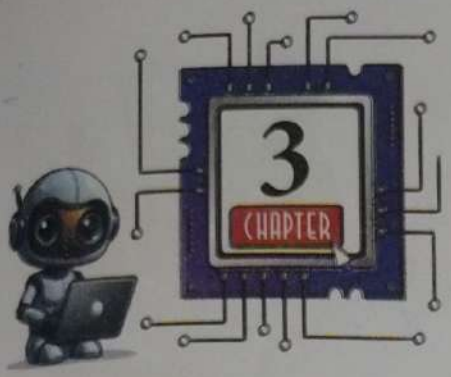
CPU	Rat	Monitor	Pen
Keyboard	Mouse	Blackboard	Phone

VI. Answer in one word :

1. Which device is used to draw pictures on the computer?
2. Which device is used to type a poem?
3. Which is the most important part of a computer?
4. Which part of the computer can be used to listen to music apart from speakers?

VII. Help the mouse to reach to the computer :





Need for a Computer

Let's ReCAll

Write the names of the parts of computer given below:






In this Chapter



- Uses of computers
- Places where computers are used

As we have learnt, computer is a smart and electronic machine. It is an important part of our life. It is used for multiple purposes.

USES OF COMPUTERS

You can chat with your friends and can send and receive E-mails.



Google

You can get information of anything by using Google.





You can play games on the computer.



You can draw and colour pictures.

You can listen to music and watch movies and cartoons.





You can buy toys and dresses online.

You can use computer for typing letters and other documents.





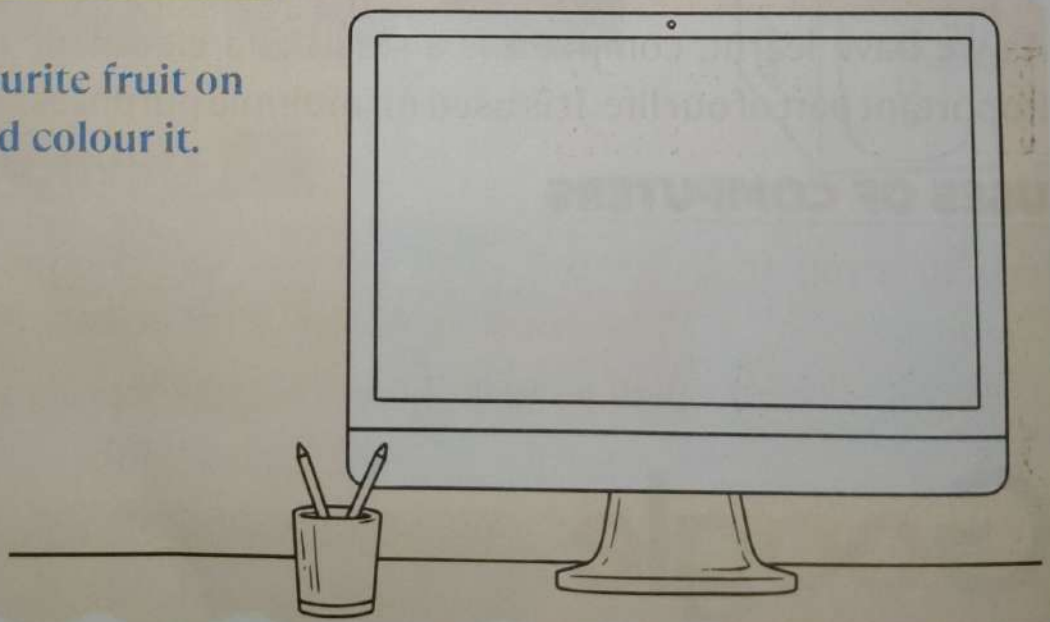
You can solve sums on the computer.

Let's Get Better

How can using a computer help you solve problems and make your life easier.

Activity Corner

Draw your favourite fruit on the monitor and colour it.



PLACES WHERE COMPUTERS ARE USED



Hospital : To Monitor the patient's health condition.



Railway Station : To book the tickets.



Office : To do the office work.



Shopping Mall : To make the bills and keep other records.



Restaurant : To keep a record of customer orders.



Bank : To deposit and withdraw the money and keep their records.



Library : To read digital books. To find books in the library.



School : To study and learn.



Police Station : To monitor the criminals through webcams.

INFO Byte

With the help of computers, we can complete some tasks in seconds and very easily.



I. Tick (✓) the correct option :

1. Which of the following activity shows solving sums on a computer?

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

2. People are using computer almost at every ..... .

(a) road ☐ (b) table ☐ (c) room ☐ (d) place ☐

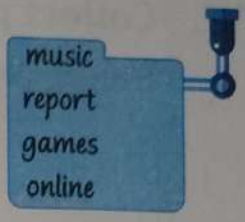
3. Which of the following activity shows working on a computer at hospital?

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

4. A computer is used to add and subtract the ..... .

(a) numbers ☐ (b) places ☐ (c) tickets ☐ (d) games ☐

II. Fill in the blanks with the help of the box :



1. We can play ..... on the computer.
2. We can take ..... video classes.
3. We can listen to ..... on speakers.
4. Teachers use computers to make test papers and ..... cards.

III. Write 'T' for True and 'F' for False :

1. A computer cannot be used in banks.
2. Computers are used at Railway stations and Airports.
3. A computer can play music.
4. You can send E-mails from a computer.

IV. Match the following :

1 

2 

3 

4 

(a) Playing a game

(b) Solving a sum

(c) Watching a cartoon film

(d) Listening to music

V. Circle the places where computers are used :

Market	Play ground	School
Railway Station	Office	Garden
Temple	Bathroom	Bank
Police Station	Hospital	Kitchen

VI. Collect pictures of any two places where computers are used. Paste them here.

Paste here

Paste here

Smart Thinking

1. What comes next in the pattern?



.....

A 

B 

C 

2. What comes next in the pattern?



.....

A 

B 

C 

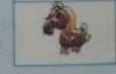
3. What comes next in the pattern?



.....

A 

B 

C 

4. Mention two pictures that come next in the pattern?

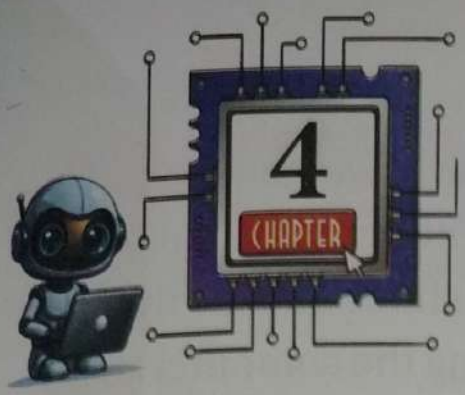


....

A 

B 

C 



Computer Etiquettes

Let's Recall

Write four things we do on a computer by taking the help of pictures given below :



1. .....



2. .....



3. .....

In this Chapter

- Computer Etiquettes
- Don'ts
- Steps to shut down a computer
- Do's
- Steps to turn on a computer

COMPUTER ETIQUETTES

Computers are very useful devices. They need to be taken care of properly. If you don't take care of them, they may not work correctly and you may not be able to complete your tasks.

To take care of them, we should follow some rules. These are called Do's. We should also avoid certain things when we are near a computer. These are called Don'ts.

Do's in the Computer Lab



Remove your shoes outside the lab before entering the computer lab.

Always enter the lab in a queue quietly.





Maintain discipline and share your computer with your friend.

Always keep your back straight. Place your feet flat on the ground. Keep your arms at the same level as the keyboard.





The top of the screen should be at your eye level. Stand up and stretch after every 20 minutes.

Always shut down the computer before leaving the lab and switch off the power supply.



Don'ts in the Computer Lab



Never talk loudly in the computer lab.



Do not get excited and start shouting by seeing any new task done by computer.



Do not take any eatables in the computer lab. Don't eat or drink near the computer.



Do not lean towards monitor. Keep distance while working on computer and don't come close to the monitor. Do not work continuously for long duration on computer.



Do not hit the mouse or keyboard too hard. Handle them gently.

Let's Get Better

Why is it important to keep the computer lab clean and follow the rules when using the computers?

HOW TO START A COMPUTER?

Starting a computer is a simple process. But you need to follow the instructions carefully otherwise your computer will not start. Here are the steps you need to follow to start the computer:

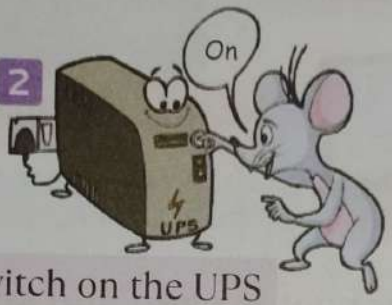
Steps to Turn On the Computer

1



Switch on the electricity for Computer.

2



Switch on the UPS

3



Switch on the Monitor.

4



Switch on the CPU.

5



Desktop will appear on the Monitor.

HOW TO SHUT DOWN A COMPUTER?

When your work is completed, you need to shut down the computer as well. Following is the process to shut down the computer:

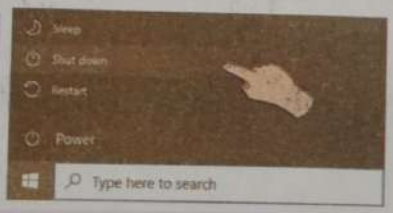
Steps to Shut Down the Computer

1



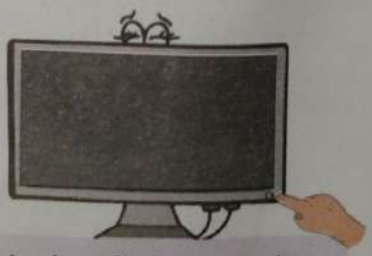
Click on Start Button on Desktop with Mouse.

2



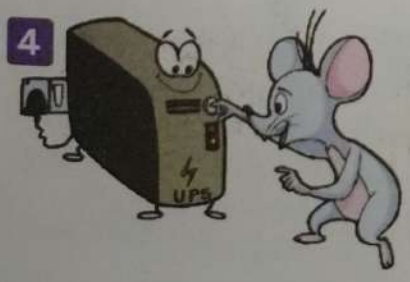
Click on Power, then click on Shut Down.

3



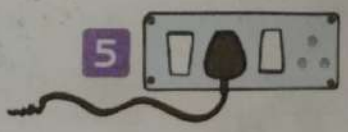
Switch off the Monitor

4



Switch off the UPS.

5



Switch off the Power switch.

Activity Corner

Colour the wrong body posture with blue and correct body posture with yellow.



Exercise

I. Tick (✓) the correct option :

1. Which of the following activity you should follow in a computer lab?

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

2. Which of the following activity you should not do in a computer lab?

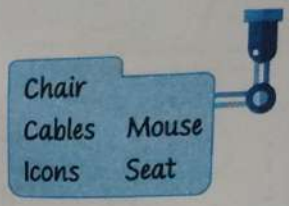
(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

3. Which of the following option shows good posture while working on a computer?

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

II. Fill in the blanks with the help of the box :

1. You should take your assigned ..... after entering the computer lab.
2. You should not play with plugs and ..... in the computer lab.
3. Press the ..... button gently.
4. Always sit straight on the ..... while working on the computer.
5. Small pictures on the Desktop are called ..... .



III. Write 'T' for True and 'F' for False :

1. You should move quietly while entering the computer lab. ☐
2. You should sleep in front of a computer. ☐
3. You should eat and drink in the computer lab. ☐
4. Start button is located on the right side corner of the Taskbar. ☐
5. Desktop screen appears when we switch on the computer. ☐

IV. Write down two Do's and two Don'ts in a computer lab :

1. .....
2. .....
3. .....
4. .....

V. Write down the steps to Turn On a computer :

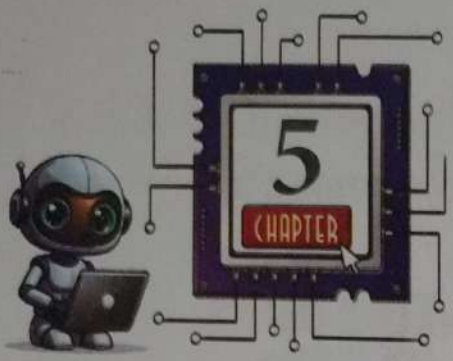
1. ..... 2. ..... 3. .....

4. ..... 5. .....

VI. Write down the steps to Shut Down a computer :

1. ..... 2. ..... 3. .....

4. ..... 5. .....



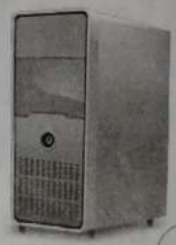
Getting Familiar with Mouse



Let's Recall

Which of the following is a mouse?





In this Chapter



- Mouse and Its uses
- Holding a Mouse
- Types of Mouse
- Working of a Mouse
- Parts of a Mouse

As we read earlier in chapter 2 that mouse is a pointing device. It is used to point and select things on the monitor.

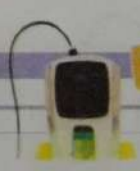


Let's get more familiar with mouse in this chapter.

MOUSE AND ITS USES

A mouse is a pointing and clicking device. It is usually connected to the computer with a wire. When you turn on the computer you will see a pointer on the monitor. It is called the mouse pointer. When you move the mouse, the pointer on the screen moves in the same direction. As you move the mouse, a red light emits from it.





A computer mouse is used for:

- Pointing at the things on the monitor.
- Selecting text, icons, files and folders.
- Drawing and colouring the pictures.
- Playing games.

TYPES OF MOUSE

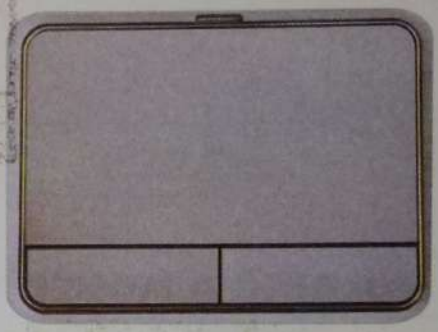
There are different types of mouse :



Wired mouse



Wireless mouse

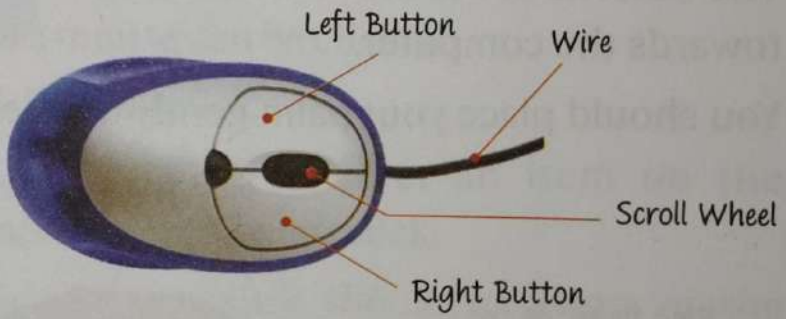


In laptops, the touchpad acts as a mouse

PARTS OF A MOUSE

A mouse has four parts :

- Left Button
- Right Button
- Scroll Wheel
- A Wire



Left Button is used to select an object or click somewhere on the computer screen. It is also used to position the cursor.

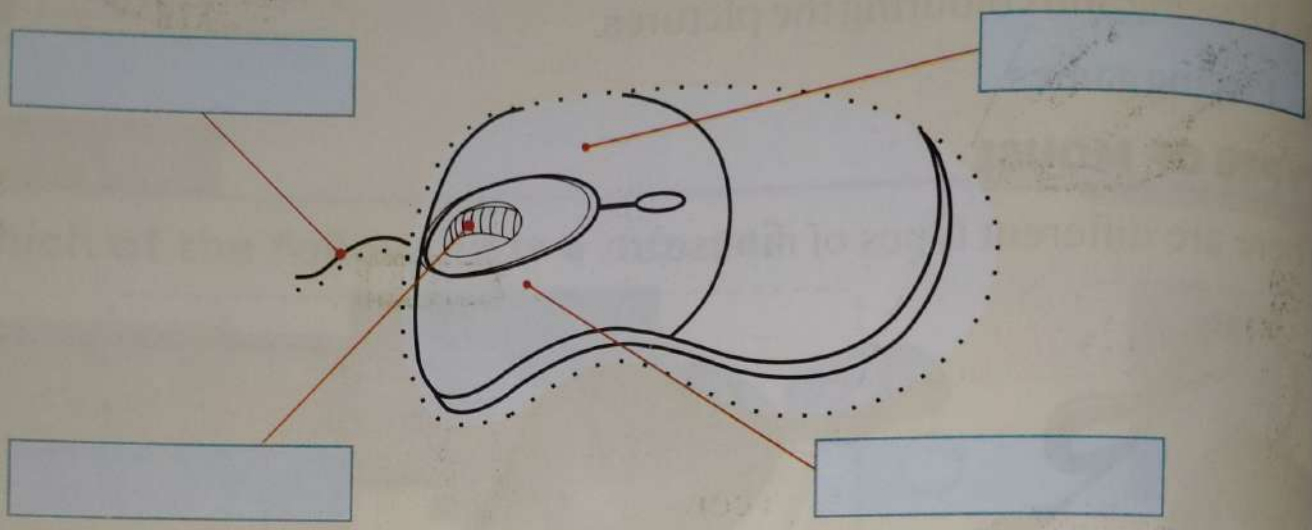
Right Button is used to open some more options for a specific task.

Scroll Wheel is used to move the pages up and down on the screen.

Wire attaches mouse to the computer through USB port.

Activity Corner

Join the dots and colour the picture. Label the parts of the mouse.



HOLDING A MOUSE

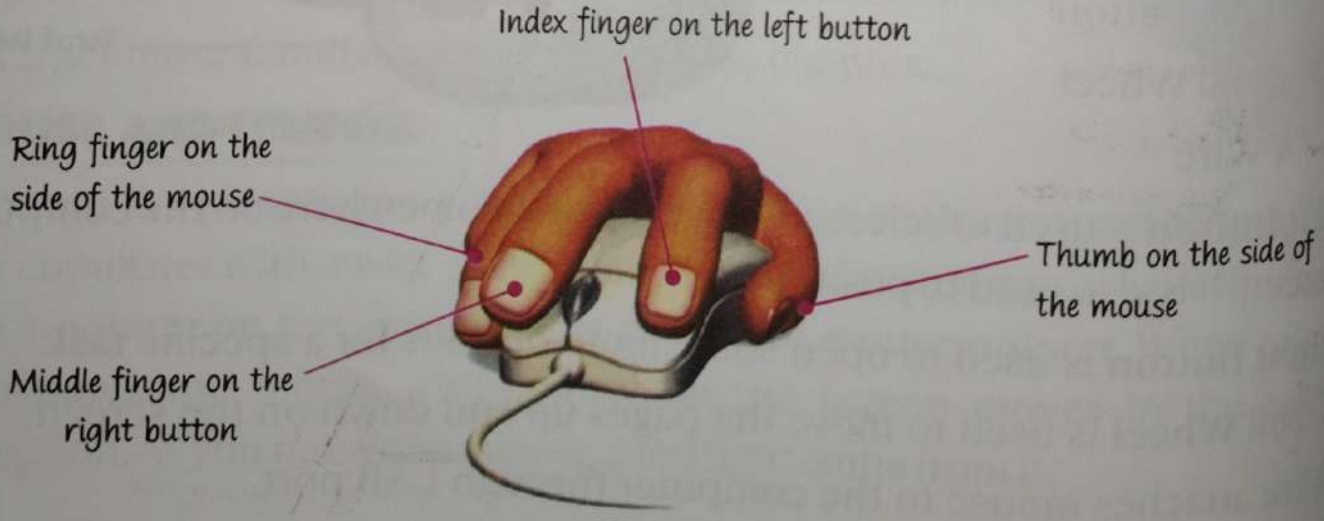
The computer mouse should be placed on a mouse pad so that it can move easily.

The buttons of the mouse should be towards the computer.

INFO Byte

When you hold a mouse, your hand should be relaxed, and your fingers should rest comfortably on the buttons. This helps you click easily.

You should place your palm gently on the mouse and hold it as shown.



If you write with your right hand, then hold the mouse in your right hand like this.



If you write with your left hand, then hold the mouse in your left hand like this.



Let's Get Better

Why is it important to hold the mouse correctly while using the computer?

Activity Corner

What is the correct way of holding a mouse? Colour the circle green for correct way red for incorrect way.



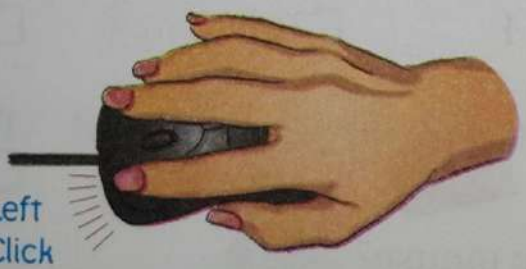


WORKING OF A MOUSE

When you press any button on the mouse, you hear a click sound. This is called pressing or clicking on the mouse.

There are different ways in which a mouse can be clicked.

1. Left click 2. Right click 3. Double click



Left-Click : To select an item on the monitor, we use left click.

Gently pressing the left button of the mouse by the index finger once and releasing it, is known as the left-click.

Right-Click : Pressing and releasing the right button of the mouse is called the right-click. Right-click helps us to see the properties of selected items.



Double-Click : Quickly pressing and releasing the left button of the mouse two times is called double-click. It helps us to open an item.





Dragging : Dragging the mouse means clicking and holding the mouse button to select an item. Then we move the mouse in desired direction while still holding the button. This action allows us to move the items on the screen. When we release the button the item is dropped in the new location.



I. Tick (✓) the correct option:

1. Identify the computer mouse.

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

2. How many buttons a mouse has ?

(a) 4 ☐ (b) 3 ☐ (c) 1 ☐ (d) 2 ☐

3. Identify the mouse pad.

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

4. Where is scroll wheel placed on the mouse?

(a) Top ☐ (b) Bottom ☐ (c) Left ☐ (d) Right ☐

II. Write 'T' for True and 'F' for False :

1. A mouse has three wires to connect the computer. ☐
2. Right button of mouse is used to open more options. ☐



3. A computer mouse rests on mouse pad.
4. Click on mouse is called shutter.

III. Fill in the blanks with the help of the box :

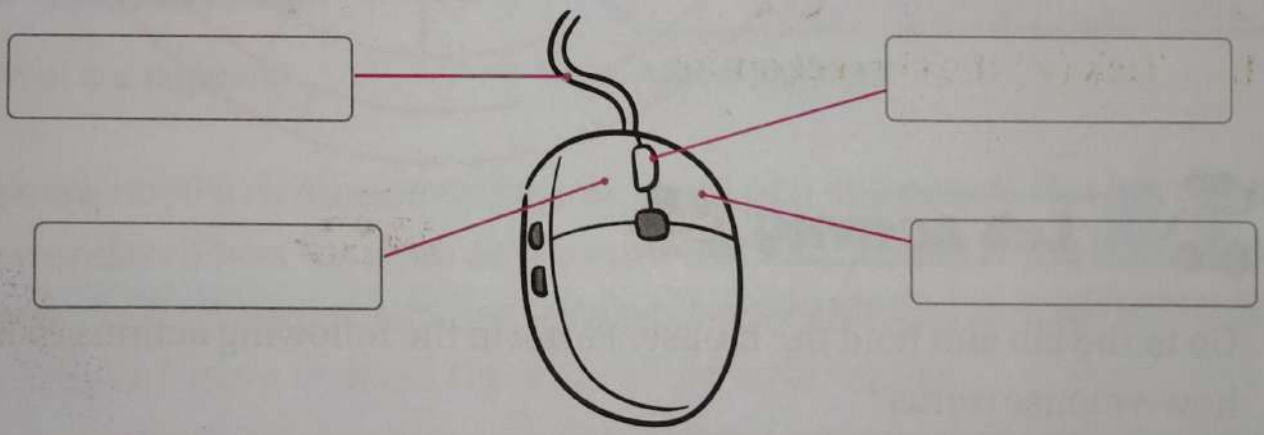
1. ..... mouse and computer mouse are different.
2. The arrow sign on the monitor is called mouse ..... .
3. When you press any button on the mouse, you hear a ..... sound.
4. The ..... of mouse should be towards the computer.

pointer real buttons click

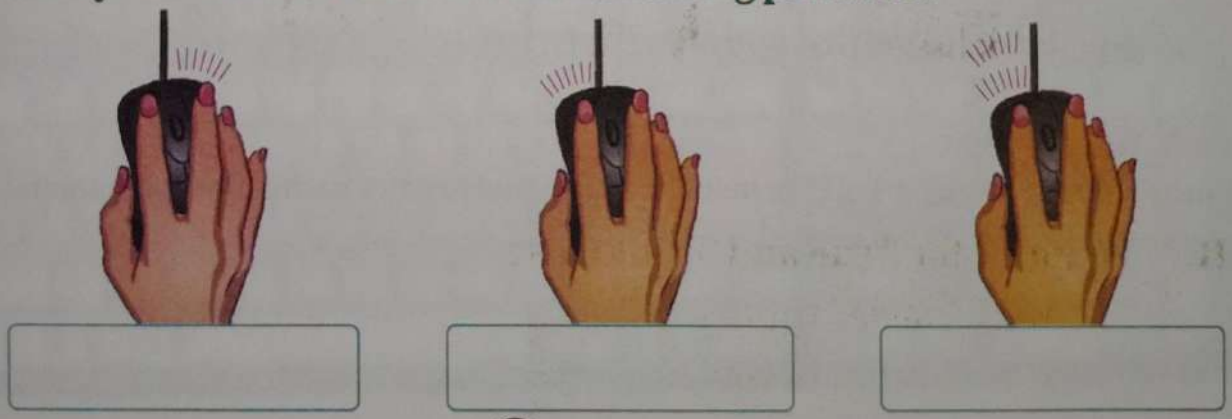
IV. Answer the following questions :

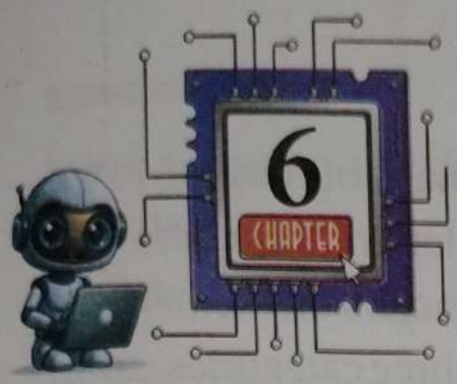
1. What is a mouse?
2. Write the name of the parts of a mouse.
3. What are single click and double click?

V. Write the name of the parts of a mouse and colour different parts with different colours :



VI. Identify the actions shown in the following pictures :





Getting Familiar with Keyboard

Let's Recall

Circle the picture that have keys.







In this Chapter



- What is a Keyboard?
- About Keys of Keyboard

When we need to write something, we use a pen or a pencil. But have you ever wondered how we write on a computer? The answer is the keyboard. We use the keyboard to write on a computer. In this chapter, we will read about the keyboard in detail and learn about its functions.





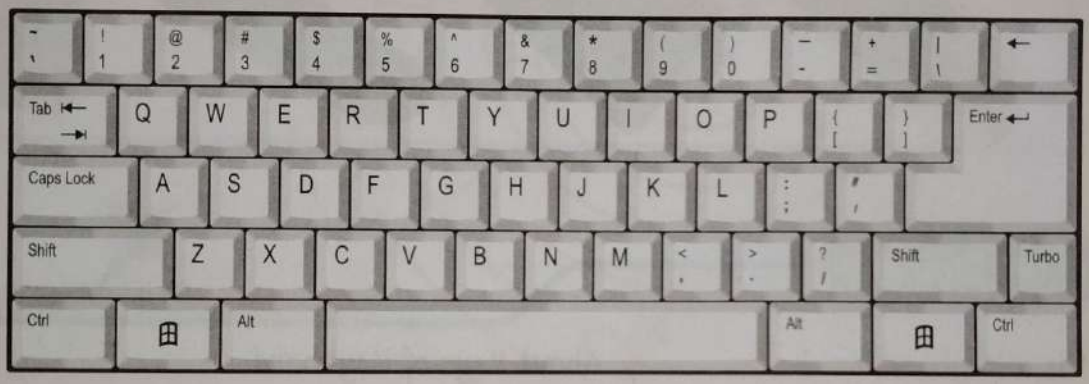
A KEYBOARD AND ITS KEYS

A keyboard is generally a rectangular board that has many buttons on it. We call them keys. It is a device that lets us to type letters, numbers and symbols into a computer. Each key has different characters on them. There are 104 keys on the keyboard. These are divided into three categories.

1. Alphabet keys 2. Number keys 3. Special keys

Activity Corner

Colour the keys which correctly spell your mother's name.



Alphabet Keys



Alphabet keys are the keys on a keyboard that have the letters A to Z. You use them to type words and sentences.

You can see that the alphabet keys are not arranged in the alphabetical order.



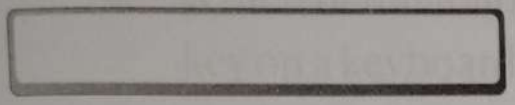
Number Keys

Number keys are the keys on a keyboard that have the numbers 0 to 9. You use them to type numbers. They are present at two places on the keyboard.

Special Keys

Special keys on a keyboard are keys that perform specific functions. They include :

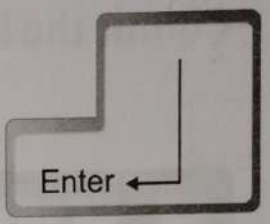
Spacebar Key



If you look at the keyboard, you will see a very long key on it in the last row. It is the spacebar key. It is used to add blank space between two letters or words.

Enter Key

Enter key is also present at two places on keyboard. It is used to start a new line or run a command.



Backspace Key



Backspace key is used to erase anything written on the left side of the cursor.

INFO Byte

When you start typing, you will see a small, blinking line or arrow on the screen that shows where you are typing or clicking. This blinking line is called cursor.

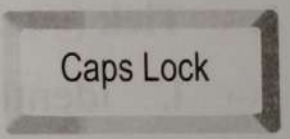


Delete Key

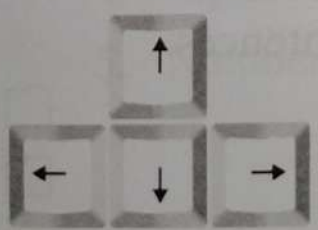
Delete key is used to erase anything that is written on the right side of the cursor.

Caps Lock Key

The Caps Lock key is a key on the keyboard that, when pressed, makes all letters you type capital until you press it again to turn it off. When Caps Lock is on, a green light on the keyboard usually turns on to indicate that it is active.



Arrow Keys



There are four arrow keys on the keyboard : Right, Left, Up and Down. They are usually located at the bottom right corner of the keyboard.

Arrow keys are keys on the keyboard that let you move the cursor or scroll the screen in four directions : up, down, left and right accordingly.

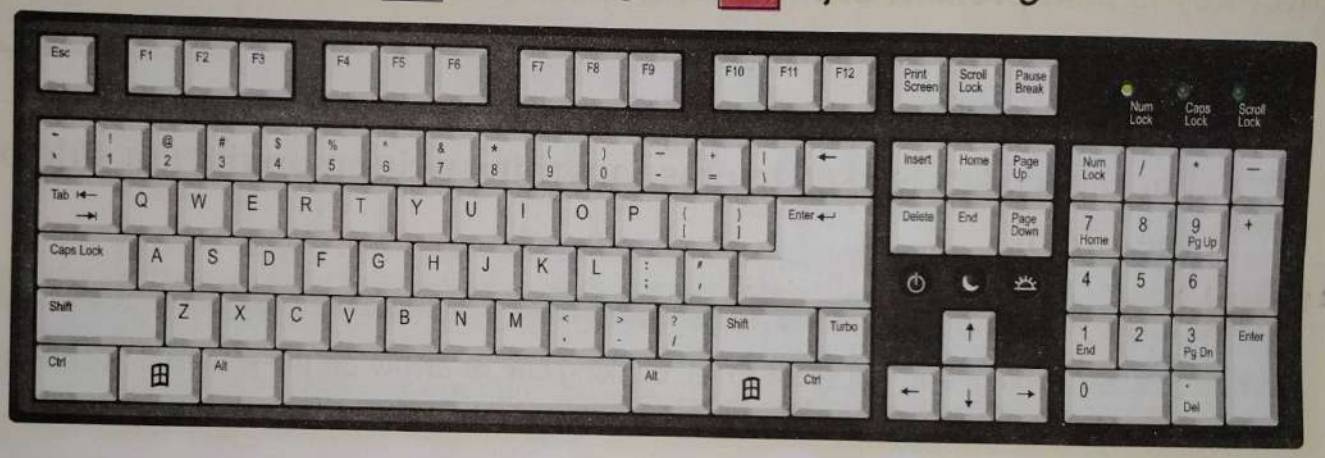
Let's Get Better

Why is it important to press the keys gently when typing on the keyboard?

Activity Corner

Colour the keys.

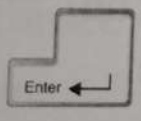
- Number keys
- Alphabet keys
- Enter keys
- Arrow keys
- Spacebar key

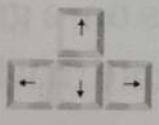
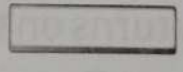


Exercise

I. Tick (✓) the correct option :

1. Identify the spacebar key.

(a) 

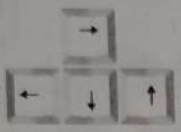
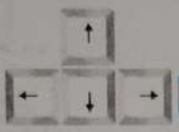
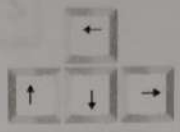
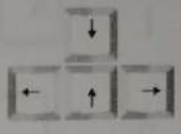
☐ (b)  ☐ (c)  ☐ (d)  ☐

2. ..... keys are used to type the words and sentences.

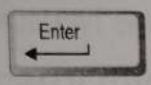
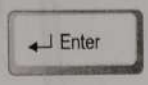
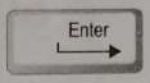
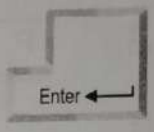
(a) Number ☐ (b) Alphabet ☐

(c) Special ☐ (d) Function ☐

3. Which of the following is the correct representation of Arrow keys on a keyboard?

(a)  (b)  (c)  (d) 

4. Which of the following illustrates the incorrect direction of Enter key on a keyboard?

(a)  (b)  (c)  (d) 

II. Write 'T' for true and 'F' for false :

1. The keyboard has number 1 to 100 on all the keys.
2. Keyboard helps to write on the Computer.
3. There are three enter keys on the keyboard.
4. There are twelve function keys on the keyboard.

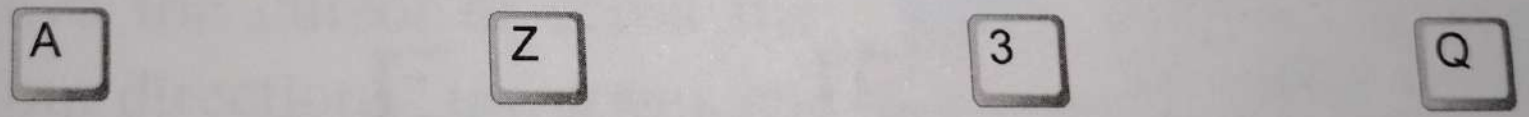
III. Answer the following questions :

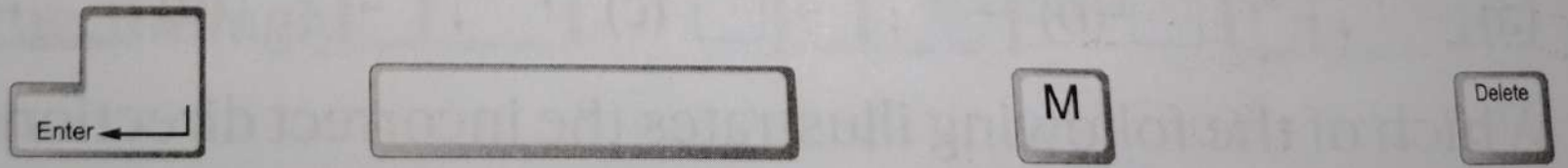
1. What is a keyboard?
2. Explain alphabet keys.
3. Write one difference between backspace key and delete key.

IV. Match the name of keys with their total number of keys :

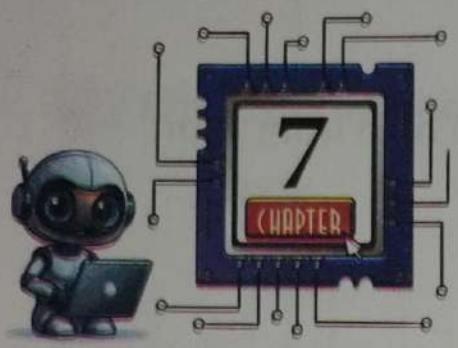
Keys	Number
1. Spacebar key	(a) 10
2. Enter keys	(b) 1
3. Delete key	(c) 2
4. Alphabet keys	(d) 1
5. Number keys	(e) 26

V. Circle the odd one out :

1. 

2. 

3. 



Hands on MS Paint



Let's Recall

Colour the circle green for the objects that you use to draw pictures.





In this Chapter



- What is MS Paint?
- Designing Tools of Paints
- Saving your Drawing
- Different Parts of Paint Window
- Shapes & Colour Pallette
- Closing the MS Paint Window

We have read earlier that a computer is used for drawing pictures and colouring item. In this chapter, we will learn how to draw on a computer.



MS PAINT

MS Paint is a computer program on your computer. It lets you to draw and colour pictures. You can use it to make drawings, edit photos and create shapes.



How to Open MS Paint?

Step 1. Click on Start button.

Start button is the icon you see on the left lower corner of your computer screen. When you click on start menu, a list of all programs will appear.

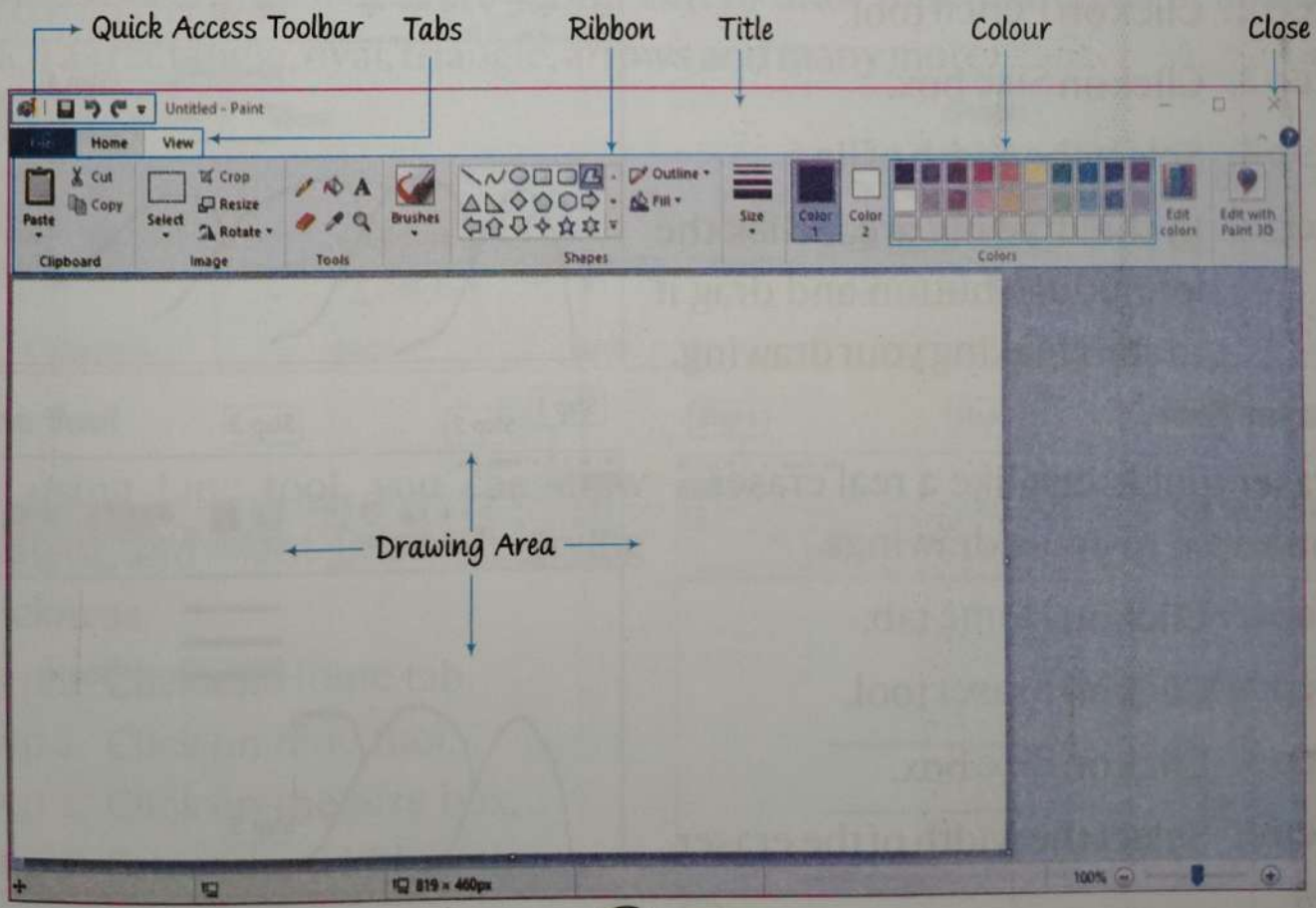
Step 2. Scroll the menu and click on Windows Accessories.

Step 3. Click on Paint.

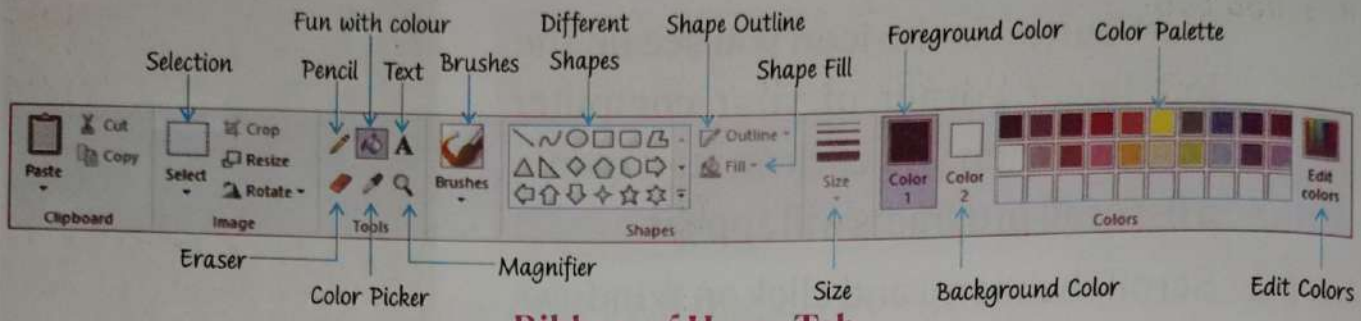


DIFFERENT PARTS OF PAINT WINDOW

Tabs : There are two tabs in MS Paint : Home and View. They contain different commands. When you click on any tab, a ribbon appears on screen.



Ribbon : A ribbon contains different groups of different tools. We use these tools to draw and colour.



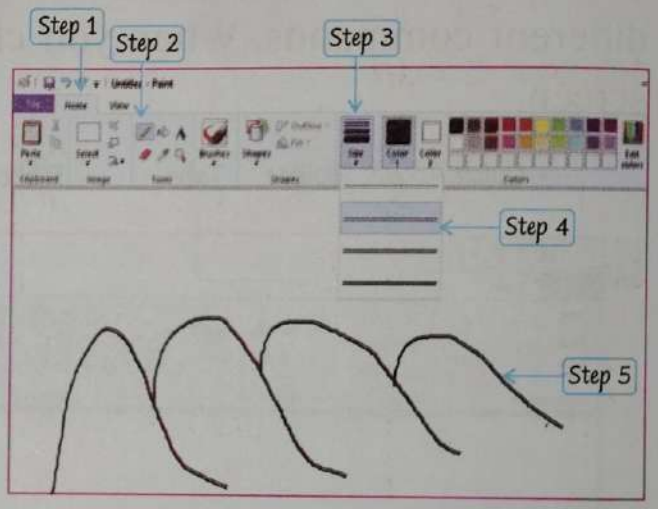
Ribbon of Home Tab

Now, let's start drawing and understand the functions of each tool.

Tools in the Tools Group

Pencil Tool

The pencil tool in MS Paint works like a real pencil. You can use it to draw straight line, wavy line or curves.



Step 1. Click on Home tab.

Step 2. Click on Pencil tool.

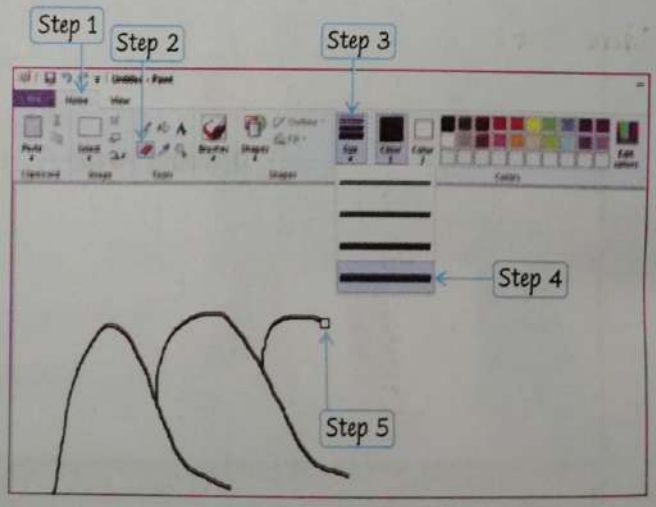
Step 3. Click on Size box.

Step 4. Select the width of line.

Step 5. In the drawing area, click the left mouse button and drag it to start making your drawing.

Eraser Tool

Eraser tool is just like a real eraser. You use it to erase drawings.



Step 1. Click on Home tab.

Step 2. Click on Eraser tool.

Step 3. Click on Size box.

Step 4. Select the width of the eraser.

Step 5. Drag it on your drawing in the drawing area to erase the required part of the drawing.

Brushes Tool

Brushes tool is used just like a real paint brush.

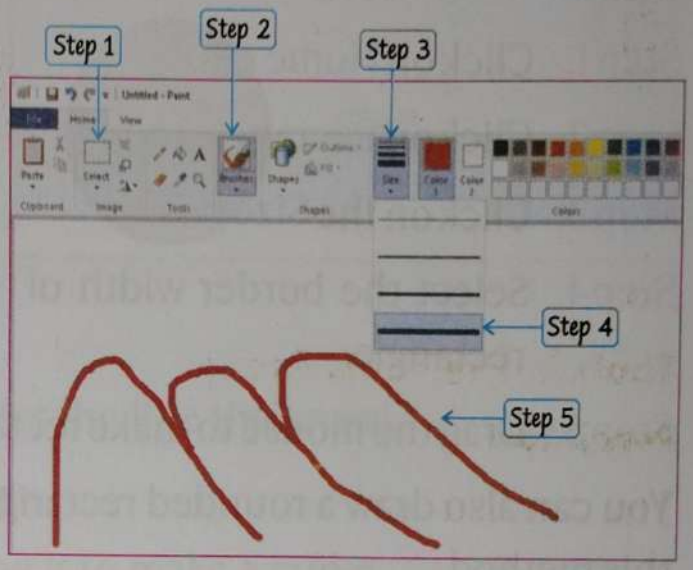
Step 1. Click on Home tab.

Step 2. Click on Brushes.

Step 3. Click on the Size box.

Step 4. Select the width of the brush.

Step 5. Now in the drawing area, drag the mouse and start making your drawing.



Shapes Group

You can use the Shapes group in Paint to draw different types of shapes such as rectangle, oval, triangle, arrows and many more.



Line Tool

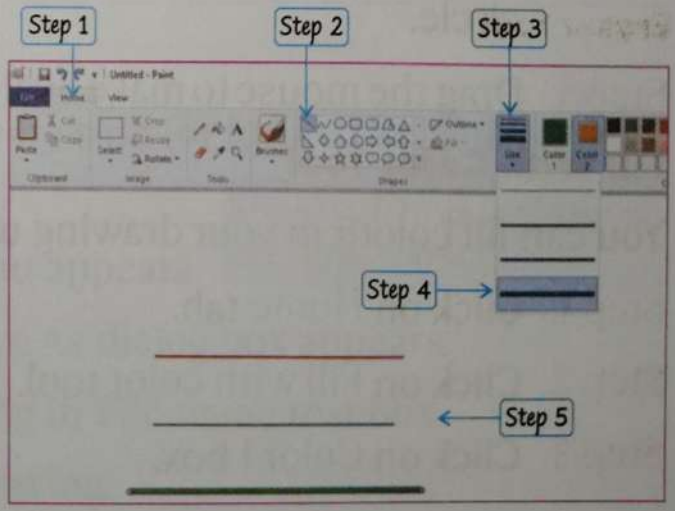
By using Line tool, you can draw straight and slant lines of varying thickness.

Step 1. Click on Home tab.

Step 2. Click on Line tool.

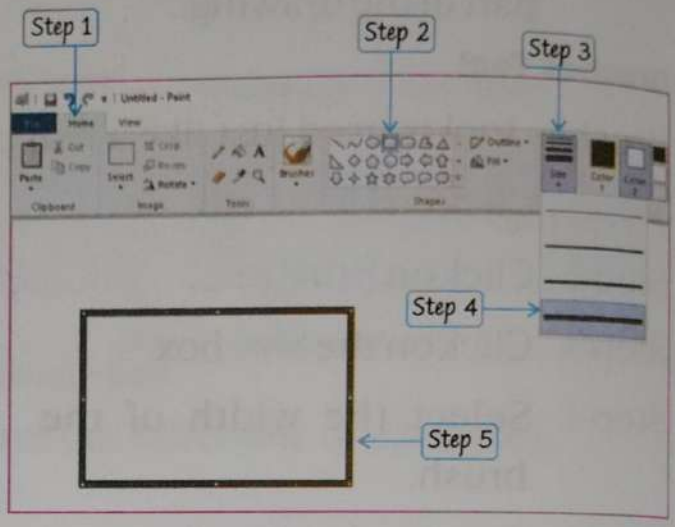
Step 3. Click on the Size box.

Step 4. Select the width of line.



Step 5. Drag the mouse to draw the line in any direction.

Rectangle Tool



You can draw rectangle and square with Rectangle tool.

Step 1. Click on Home tab.

Step 2. Click on Rectangle tool.

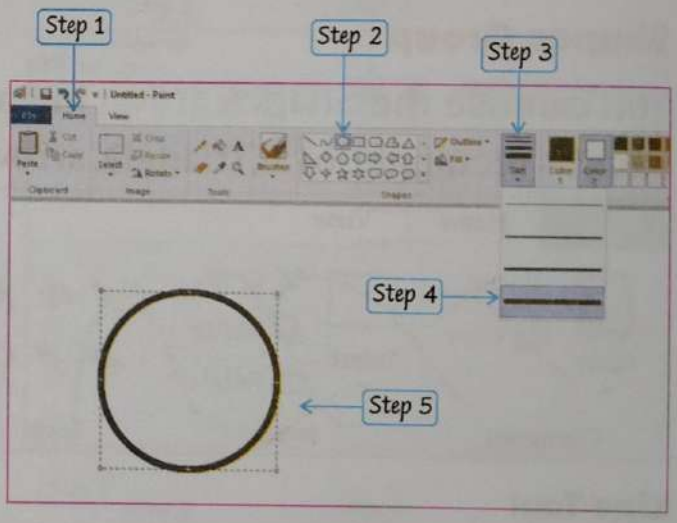
Step 3. Click on the Size box.

Step 4. Select the border width of rectangle.

Step 5. Drag the mouse to make rectangle.

You can also draw a rounded rectangle, triangle and other shapes by using this method.

Oval Tool



Oval tool is used to draw circle and oval shapes.

Step 1. Click on Home tab.

Step 2. Click on Oval tool.

Step 3. Click on the Size box.

Step 4. Select the border width of circle.

Step 5. Drag the mouse to make oval or circle.

Fill With Color Tool

You can fill colour in your drawing using Fill With Color tool.

Step 1. Click on Home tab.

Step 2. Click on Fill with color tool.

Step 3. Click on Color 1 box.

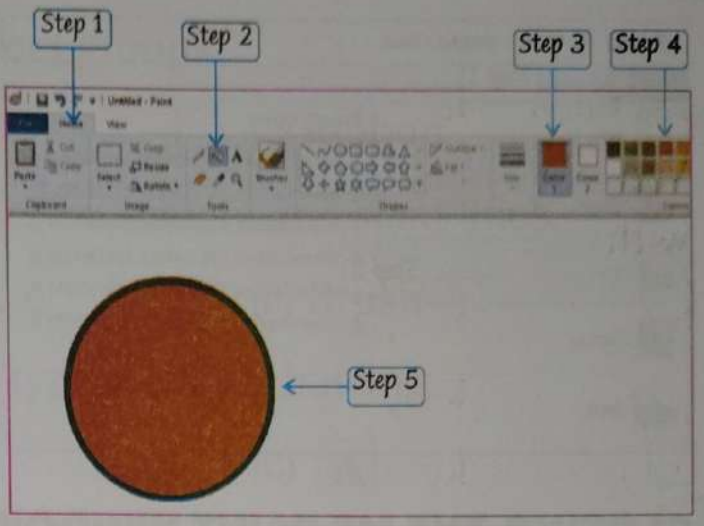
INFO Byte

You can pick from many colours to colour your drawing.



Step 4. Click on any colour from the Color group.

Step 5. Click inside the area where you want to colour.



Curve Tool

You can draw curved lines with Curve tool.

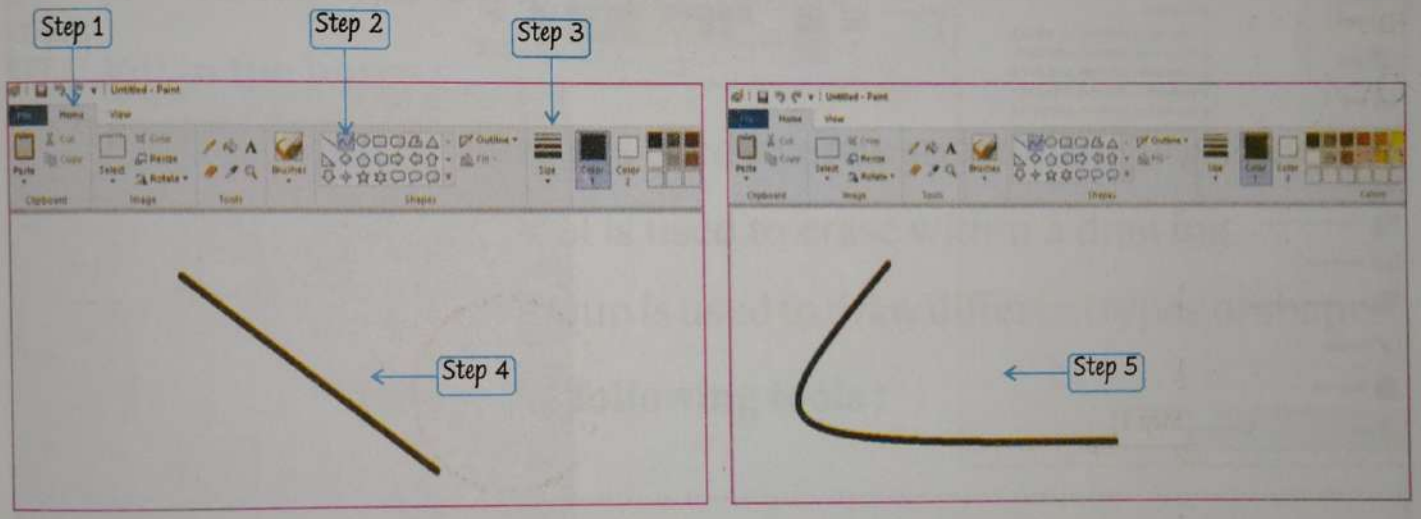
Step 1. Click on Home tab.

Step 2. Click on Curve tool.

Step 3. Click on the Size box and select the line thickness.

Step 4. Drag the mouse to make a line.

Step 5. Now, click on the line and drag it to make a curve.



SAVING YOUR DRAWING

While drawing, you must save it side-by-side for future use. Follow these steps to save your drawing.

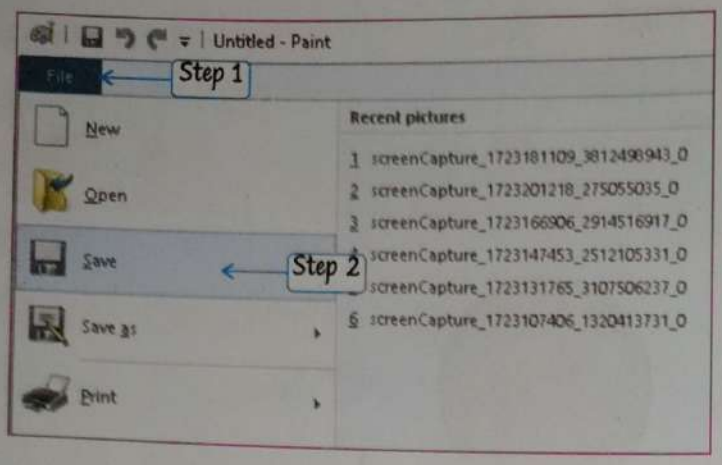
SHORTCUT SECTION To save a drawing, press Ctrl + S

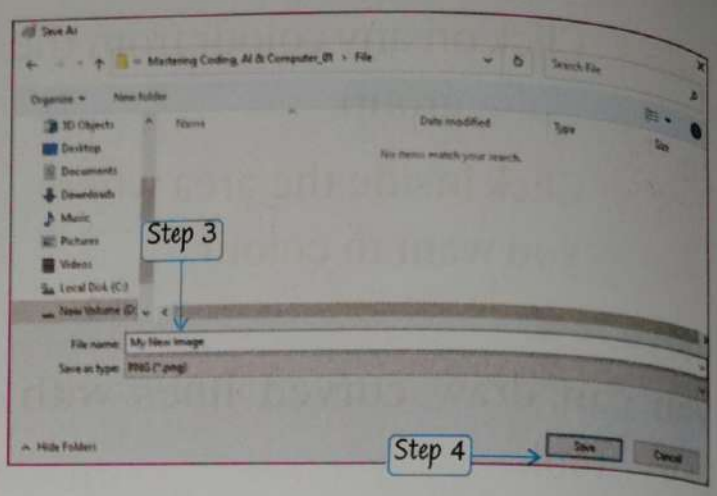
Step 1. Click on File option. File menu appears.

Step 2. Click on Save option. The Save As dialog box appears.

Step 3. Type the name for the drawing in File name text box.

Step 4. Click on Save to save your drawing.





CLOSING THE MS PAINT WINDOW

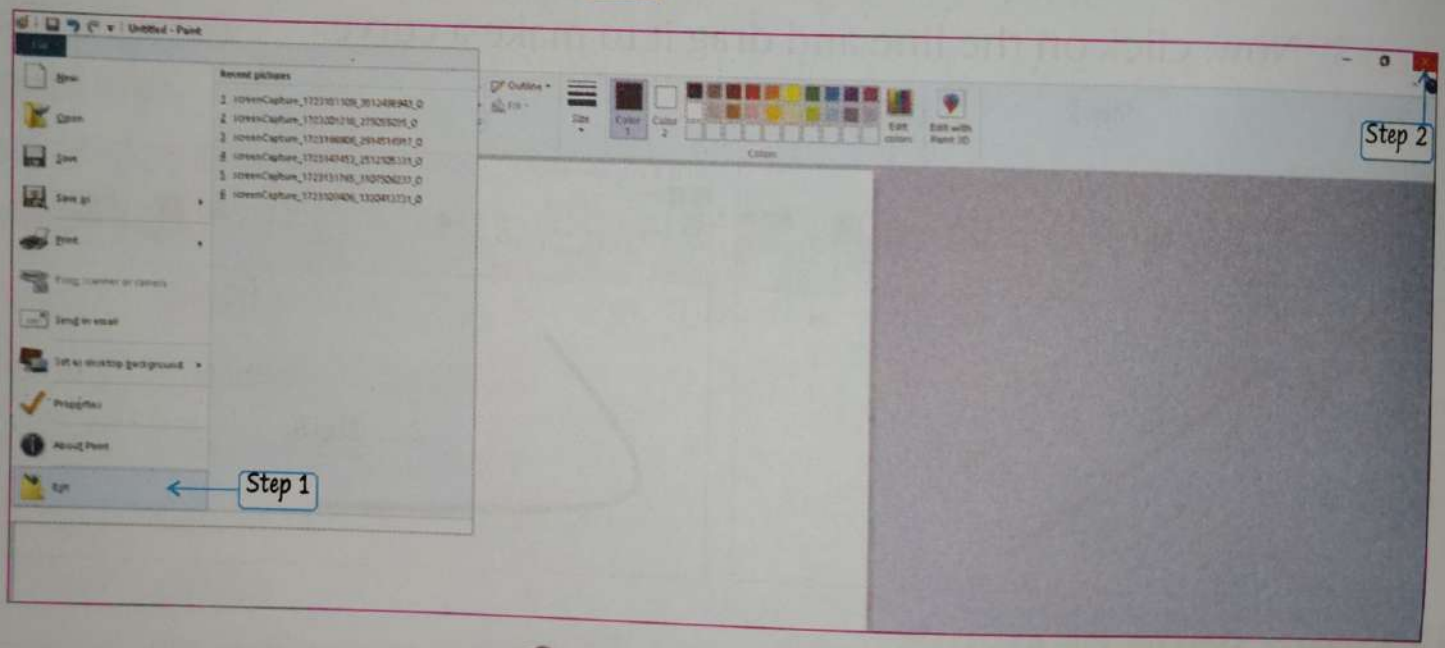
To close paint, follow these steps.

Step 1. Click on File tab.

Step 2. Click on Exit or close ✕ sign.

Let's get BETTER

How can using Paint help you express your ideas and creativity?





I. Tick (✓) the correct option :

1. ..... program is used to draw and colour objects.

(a) Paint ☐ (b) Notepad ☐ (c) Calculator ☐

2. ..... contains different groups.

(a) Canvas ☐ (b) Tab ☐ (c) Ribbon ☐

3. ..... tool is used to draw straight and slant lines.

(a) Line ☐ (b) Circle ☐ (c) Brush ☐

4. ..... tool is used to type within a drawing.

(a)  ☐ (b)  ☐ (c)  ☐

II. Write 'T' for True and 'F' for False statements :

1. Colors group is at the left side of the ribbon. ☐
2. Oval tool is used to delete a drawing. ☐
3. Curve tool is used to draw straight lines. ☐
4. We can type text in Paint program. ☐

III. Fill in the boxes :

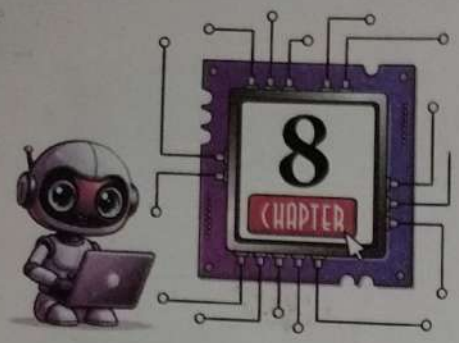
1. P _ N _ _ L tool is used to draw straight and wavy lines.
2. E _ _ S _ R tool is used to erase within a drawing.
3. S _ A _ _ S group is used to draw different types of shapes.

IV. Write the names of the following tools :

1.  .....
2.  .....
3.  .....

V. Answer the following Questions :

1. What is MS Paint?
2. Write the name of different Parts of Paint Window.
3. What are a line and an oval?



Sequencing and Patterns

Let's Recall

Complete the following series:

(i) A C E G ....

(ii) 2 4 6 8 ....

In this Chapter



- What is a sequence?
- What is a pattern?
- What is a loop?

WHAT IS A SEQUENCE?

A sequence is a list of things in a specific and correct order. For example: We all know that there are seven days in a week. They are:

MONDAY TUESDAY WEDNESDAY

THURSDAY FRIDAY SATURDAY

SUNDAY

The days of a week form a sequence. They always come in the same order. First day of the week will always be Monday, then Tuesday and so on.



Now, let us look at a story given below:

The Story of Thirsty Crow

1. 

A crow was looking for water.

2. 

He found the water in a vessel.

3. 

But the level of water was down.

4. 

He smartly picked the stones nearby.

5. 

And put them in the water to make water level rise.

6. 

After putting many stones, he was able to drink water.

If we think about the above story, we will find that the crow followed some steps to drink the water.

What do you think, if crow had skipped any one step, would it have been able to drink the water?

The steps that the crow followed to drink water are called a Sequence.

Activity Corner

Look at the pictures of the daily routine and number them in the correct sequence.





WHAT IS A PATTERN?

A pattern is something that repeats in a regular way. Let us understand it with the help of some examples.



In the picture shown above, we can see that two shapes are used. One is star and other is circle.

Let's Get Better

Why is it important to notice patterns in the things around us, like in nature or our daily activities?

They are arranged by first putting a star and then a circle. This shows us a pattern that a star is followed by a circle and a circle is followed by a star.

WHAT IS A LOOP?

When a pattern repeats again and again, it is called a loop. For example :

Skipping the Rope

It is like jumping again and again until you decide to stop. Same as loop in coding, which repeats an action until you tell it to finish (stop).



Activity Corner

Is this a Loop? Put a tick (✓) mark if YES or a cross (✗) mark if NO.





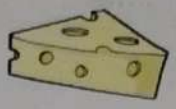


Exercise

I. Tick (✓) the correct option :

1. What is a sequence?
   - (a) A kind of story
   - (b) A list of actions
   - (c) A type of printer
   - (d) None of these
2. What is a pattern?
   - (a) Repetition in a regular way
   - (b) A star
   - (c) A circle
   - (d) All of these
3. What is a loop?
   - (a) A list of actions
   - (b) A pattern
   - (c) A sequence
   - (d) A repeated pattern

II. Choose the correct sequencing :

1. (a)  (b)  (c)  (d) 

(i) (a), (c), (d), (b) ☐ (ii) (d), (a), (c), (b) ☐

(iii) (a), (d), (c), (b) ☐ (iv) (a), (d), (b), (c) ☐

2. (a)  (b)  (c)  (d) 

(i) (b), (c), (d), (a) ☐ (ii) (c), (b), (d), (a) ☐

(iii) (b), (c), (a), (d) ☐ (iv) (c), (b), (a), (d) ☐

3. (a)  (b)  (c)  (d) 

(i) (b), (a), (c), (d) ☐ (ii) (a), (c), (b), (d) ☐

(iii) (b), (c), (d), (a) ☐ (iv) (a), (b), (c), (d) ☐

III. Match the following :

1.  (a) Loop
2. Weekdays (b) pattern
3. Skipping (c) Loop
4. Cycling (d) Sequencing

IV. Define :

1. Sequencing

.....

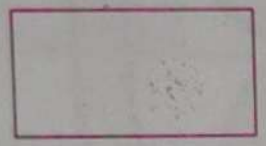
2. Loop

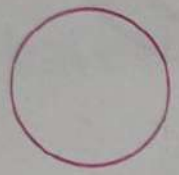
.....

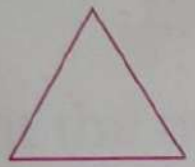
3. Pattern

.....

V. Match the shapes and colour them :

1. 

2. 

3. 

(a) 

(b) 

(c) 



Introduction to AI and Robotics



Let's Recall

(i) If Rohan has 3 toffees and Mohan has 5 toffees, how many toffees do they have in total?



(ii) Count the number of fruits in the basket and write their names along with how many of each are there.

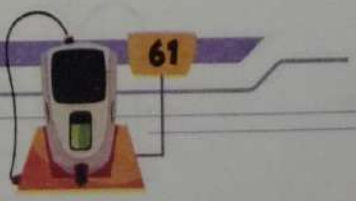




In this Chapter



- What is Intelligence?
- Concept of AI
- Concept of Robot



Intelligence is the ability to learn, understand and solve the problems. It helps you to think, reason and make decisions. Let us understand this with the help of an example :



Akash is standing on a traffic signal.
The signal shows the red light.

Should he cross the road or not?

Yes ☐ No ☐

This ability to think about a situation and make correct decision is called intelligence.

ARTIFICIAL INTELLIGENCE

We have the ability to think and make decisions. Similary, if machines can learn and make decisions on it's own then it is called Artificial Intelligence.

Let us look at some examples of artificial intelligence :

Smart phones have voice assistants like siri and google assistant. You can talk to them. They will answer your questions and help you with tasks.



Voice Assistant



Smart Watch

Now-a-days, we have smart watches also. They perform various tasks like tracking and counting your steps and many more.

Smart Camera is an intelligent device. It can detect any movement and can rotate automatically.



Smart Camera

Smart doors open and close automatically, when someone comes near the door. They senses movement to do so.



Smart Doors



Smart Speaker

You can communicate with a smart speaker. It listens to the command when you speak and responds. You can hear songs on it.

Smart cars run automatically without any driver. They are safe and useful.



Smart Car

WHAT IS A ROBOT?

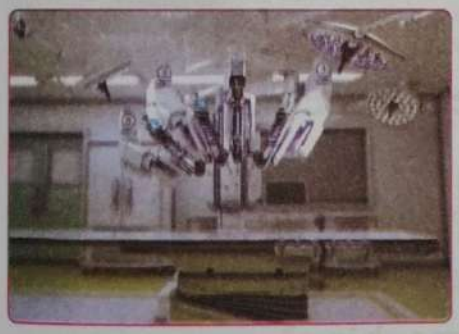
A robot is a machine that can do tasks on it's own. It can move, sense and perform actions.

The brain is more powerful than any intelligent machine. Intelligent machines like robot are made by man. Let us look at some types of robots.

Smart robot can play music, dance and tell stories.

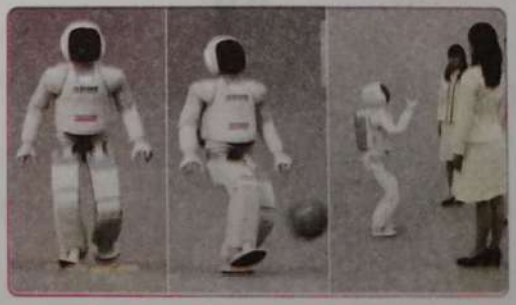
Roomba is a robotic vacuum cleaner. It cleans the floor of a house automatically.





Da Vinci Surgical system robot is used in hospitals to help doctors in operations.

This is ASIMO robot. It can walk, talk and interact with people.



These are flying robots used for taking photos, delivering packages, etc. They are called drones.





This is a robotic dog called AIBO. It is a smart toy. It can play and talk with you.

Let's Get Better

How can robots and AI help us in our daily lives?



I. Tick (✓) the correct option :

1. Which of the following has natural intelligence?

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

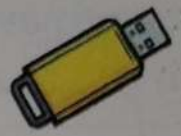
2. Which of the following has artificial intelligence?

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

3. Which of the following is an intelligent machine?

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

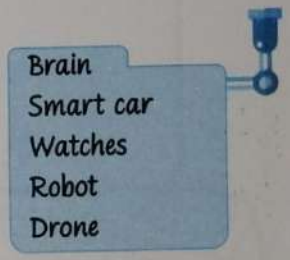
4. Which of the following is an intelligent device?

(a)  ☐ (b)  ☐ (c)  ☐ (d)  ☐

II. Match the following :

1.  •		• (a) Flying Robot
2.  •		• (b) Smart Camera
3.  •		• (c) Toy Robot
4.  •		• (d) Smart Speaker

III. Fill in the blanks with the help of the box :



1. On smart ..... you can dial or receive the calls.
2. We use our ..... and intelligence to find the answer.
3. ..... camera is called flying robot.
4. ..... runs without driver.
5. ..... needs orders or commands from humans to do any tasks.